MTH1004

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2018

Probability, Statistics and Data

Module Leader: Dr Christopher A. T. Ferro

Duration: 2 HOURS.

The mark for this module is calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Graph Paper and Statistical Tables (Murdoch & Barnes) will be provided. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Among a group of 40 students, 25 have mobile phones, 22 have tablets, and 17 have laptops. Moreover, 11 have both phones and tablets, 10 have both phones and laptops, and 8 have both laptops and tablets. How many students have all three? (5)

   (b) Let X be a discrete random variable with probability mass function $p(x) = P(X = x)$.

   (i) What properties must $p(x)$ satisfy?

   (ii) Define $E\{g(X)\}$, the expectation of a real-valued function, $g(X)$.

   (iii) Let $g(X) = aX + b$ for real constants a and b. Show that

   $$E\{g(X)\} = aE(X) + b.$$

   (9)

   (c) The times taken for 10 air-conditioning units to fail were 143, 164, 188, 188, 190, 192, 206, 209, 213 and 216 hours. The mean of this sample is 190.9 minutes. Consider modelling these data as realisations of i.i.d. random variables, $Y_i \sim \text{Exp}(\theta)$ for $i = 1, \ldots, 10$ with probability density function

   $$f(y; \theta) = \theta e^{-\theta y} \qquad \text{for } y > 0.$$

   (i) Explain briefly whether the Exponential distribution is a sensible model for these data.

   (ii) Show that $E(Y_i) = 1/\theta$ and hence compute the method of moments estimate of θ.

   (iii) Describe how you would use simulation to estimate the standard error for this estimate. (19)

   (d) (i) Explain what is meant by type I and type II errors in hypothesis testing.

   (ii) The level of the River Exe varies naturally according to a Normal distribution with mean 0.5 metres and standard deviation 0.2 metres. A path is submerged if the river level exceeds 0.75 metres. Use the statistical tables to compute the proportion of time that the path is submerged.

   (iii) The river level was measured once a day for 20 days and the sample mean level was 0.6 metres. Use the statistical tables to conduct a test of size 5% of the null hypothesis that the mean river level is 0.5 metres against the one-sided alternative that it is higher than that. State clearly your test statistic, its null distribution, the critical region and your conclusion. (17)

[**50**]

SECTION B

2. Barry from Bristol runs Tamar.co.uk, an internet shopping site serving the South West. When he can, he personally delivers items to their destination from his warehouse in Bristol at negligible cost. His acquaintance Eddie from Exeter has his own company and warehouse, and Barry will pay Eddie to deliver items from Exeter in situations where he cannot personally deliver on time. In these cases, Eddie has a flat fee of £10 if the return journey is expected to take less than 3 hours, and charges a further £10 thereafter.

   There are 2 main driving routes from Bristol to Exeter, both take approximately 90 minutes. There are 2 routes from Exeter to Truro, one over Dartmoor that takes approximately 80 minutes, the other via Plymouth takes 150 minutes.

   During a storm, trees fall and block any given road independently with probability p. Blocked roads are immediately closed by the police. When the storm is over and before any clear up operation has time to begin, Barry takes a large order to be delivered immediately to Truro.

   (a) What is the probability that Barry cannot personally deliver the order? (9)

   (b) What is the probability that, in the case Barry cannot deliver the order, he is able to hire Eddie to deliver for him? You may assume Eddie is available and has the required items in his own warehouse. (10)

   (c) If Barry cannot deliver the order, but Eddie can, what is the probability that it will cost him more that £10? (6)

   [**25**]

3. Let X_1 and X_2 be independent $\mathrm{N}(0,1)$ random variables with probability density function

$$f(x) = (2\pi)^{-1/2} e^{-x^2/2} \qquad \text{for } -\infty < x < \infty$$

and let $Y_1 = X_1/X_2$.

(a) Write down the range space of Y_1 and show that its probability density function is

$$f_{Y_1}(y) = \frac{1}{\pi(1+y^2)}. \tag{19}$$

(b) Given that

$$\frac{\mathrm{d}}{\mathrm{d}y} \tan^{-1}(y) = \frac{1}{1+y^2},$$

show how the probability integral transform can be used to simulate from the distribution of Y_1. (6)

[**25**]

4. The 1986 crash of the space shuttle Challenger was linked to failure of O-ring seals. The number of O-ring failures was recorded for each of 23 previous shuttle missions. Each shuttle has 6 O-rings. The sample mean number of failures per mission was 0.478 and the sample variance was 1.119.

(a) Use these data, a parametric statistical model and the method of moments to estimate the probability that at least one O-ring failed on the Challenger shuttle. State clearly any assumptions you make. (15)

(b) Explain what is meant by a good model fit. Name one computational and two graphical ways that you would use to check whether your model is a good fit. Perform the computational check on your model and then comment briefly on the fit of the model and hence on the validity of your result in part (a). (10)

[**25**]